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Machine Learning MDS504 – 3

Topic: Implementation of the Research article “Fashion Images Classification using Machine Learning, Deep Learning and Transfer Learning Models”

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1. Introduction

Machine learning and deep learning techniques have become fundamental tools for solving complex image classification problems. In recent years, researchers have explored various approaches to improve classification accuracy and computational efficiency in visual recognition tasks. One commonly used benchmark dataset for evaluating image classification algorithms is the Fashion-MNIST dataset, which contains grayscale images representing different clothing categories.

The research article titled *“Fashion Images Classification using Machine Learning, Deep Learning and Transfer Learning Models”* investigates the performance of multiple machine learning and deep learning algorithms on the Fashion-MNIST dataset. The study compares classical machine learning models such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest, and Decision Tree with deep learning architectures including Convolutional Neural Networks (CNN) and other advanced models. The paper also analyzes the effect of dimensionality reduction using Principal Component Analysis (PCA) on model performance.

In this implementation project, the methodology described in the selected research article was reproduced using the Fashion-MNIST dataset. Several machine learning models were implemented with and without PCA-based dimensionality reduction, and their performance was compared. In addition, a convolutional neural network (CNN) was implemented to evaluate deep learning performance on the same dataset. The obtained results were analyzed and compared with those reported in the research paper.

2. Dataset Used and Selected Model Architecture

The dataset used in this implementation is the **Fashion-MNIST dataset**, which is a widely used benchmark dataset for image classification tasks. Fashion-MNIST was developed as a replacement for the traditional MNIST handwritten digit dataset and consists of grayscale images representing different clothing items. The dataset contains **70,000 images**, divided into **60,000 training images** and **10,000 testing images**. Each image has a resolution of **28 × 28 pixels**, resulting in **784 pixel values per image**. The images belong to **10 different classes**, which represent various types of clothing items. Some of the classes included in the dataset are Trouser, Sandal, Bag, Pullover, Dress etc. Each pixel value represents grayscale intensity ranging from **0 to 255**. Before training the models, the pixel values were normalized to improve model training stability.

For machine learning models, each image was flattened into a **784-dimensional feature vector**. To analyze the effect of dimensionality reduction, **Principal Component Analysis (PCA)** was applied to reduce the number of features from **784 to 340 components**, as described in the selected research article. For deep learning experiments, a **Convolutional Neural Network (CNN)** architecture was implemented. CNN models are well suited for

image classification tasks because they automatically extract spatial features from images using convolutional filters.

The implemented CNN architecture consists of convolutional layers for feature extraction, max-pooling layers for dimensionality reduction, dropout layers for preventing overfitting, and fully connected dense layers for classification. The final output layer uses a softmax activation function to classify the images into the ten clothing categories.

3. Mathematical Background

Machine learning and deep learning models rely on mathematical techniques to extract patterns from data and make predictions. In this implementation, several algorithms were used including Principal Component Analysis (PCA), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest, and Convolutional Neural Networks (CNN).

Principal Component Analysis (PCA)

PCA is a dimensionality reduction technique used to transform high-dimensional data into a smaller set of features while preserving most of the variance in the data. PCA works by computing the eigenvectors and eigenvalues of the covariance matrix of the dataset. The eigenvectors represent the principal components, which are directions in which the data varies the most. Mathematically, PCA transforms the original data matrix X into a new feature space:

$$Z = XW$$

where W represents the matrix of principal components. In this implementation, PCA was used to reduce the number of features from **784 pixels to 340 principal components**, which helps reduce computational complexity while retaining important information.

Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning algorithm used for classification tasks. The main objective of SVM is to find the optimal hyperplane that separates different classes in the feature space.

The decision boundary is defined as:

$$w \cdot x + b = 0$$

where, w is the weight vector, x is the input feature vector, and b is the bias. SVM attempts to maximize the margin between the hyperplane and the nearest data points (called support vectors), which improves the generalization ability of the model.

K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a simple non-parametric classification algorithm that classifies a data point based on the majority class of its nearest neighbors in the feature space. The distance between data points is typically calculated using Euclidean distance:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}$$

The algorithm selects the **k nearest neighbors** and assigns the class that appears most frequently among those neighbors.

Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and combines their outputs to improve prediction accuracy. Each tree is trained on a random subset of the data and features, which helps reduce overfitting. The final prediction of the Random Forest model is obtained through majority voting among all the individual decision trees.

Convolutional Neural Networks (CNN)

Convolutional Neural Networks are deep learning architectures specifically designed for image processing tasks. CNNs automatically learn spatial features from images using convolutional filters. A typical CNN architecture consists of several layers including convolutional layers, pooling layers, and fully connected layers. Convolutional layers apply filters to detect important patterns such as edges and textures. Pooling layers reduce the spatial dimensions of the feature maps, while fully connected layers perform the final classification. The output layer usually uses the **softmax activation function**, which converts the network outputs into probability distributions across the different classes.

4. Libraries and Tools Used

In this implementation, Python programming language was used along with several other libraries. The experiments were performed using **Google Colab**, which provides a cloud-based environment for running Python code and training machine learning models. The following libraries were used in this implementation:

- **NumPy**

It provides support for multi-dimensional arrays and mathematical operations that are required for data preprocessing and matrix computations.

- **Matplotlib**

Matplotlib is a visualization library used to generate graphs and plots. In this project, it was used to visualize training accuracy, loss curves, and model comparison graphs.

- **TensorFlow / Keras**

TensorFlow is an open-source deep learning framework, and Keras is a high-level API used for building and training neural networks. These libraries were used to implement the CNN model for image classification.

- **Scikit-learn**

In this project, Scikit-learn was used to implement the machine learning models including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest, and Principal Component Analysis (PCA).

- **Seaborn**

Seaborn is a visualization library built on top of Matplotlib that provides improved graphical representations. It was used to visualize the confusion matrix for the classification results.

5. Training Configuration and Hyperparameters

Training configuration refers to the parameters and settings used while training the machine learning and deep learning models. Proper selection of these parameters helps improve model performance and stability.

For the **Convolutional Neural Network (CNN)** model, the images were reshaped to include a channel dimension of $28 \times 28 \times 1$, which is required for convolutional operations. The model was trained for **10 epochs** using the **Adam optimizer**, which is commonly used for deep learning tasks due to its adaptive learning rate capabilities. The loss function used for training was **Sparse Categorical Crossentropy**, which is suitable for multi-class classification problems where the labels are represented as integers. During training, the model learned to minimize the loss function while improving classification accuracy on the training dataset. To reduce overfitting, **dropout layers** were included in the CNN architecture. Dropout randomly disables a portion of neurons during training, which helps improve the generalization capability of the model.

For the machine learning models, the following configurations were used:

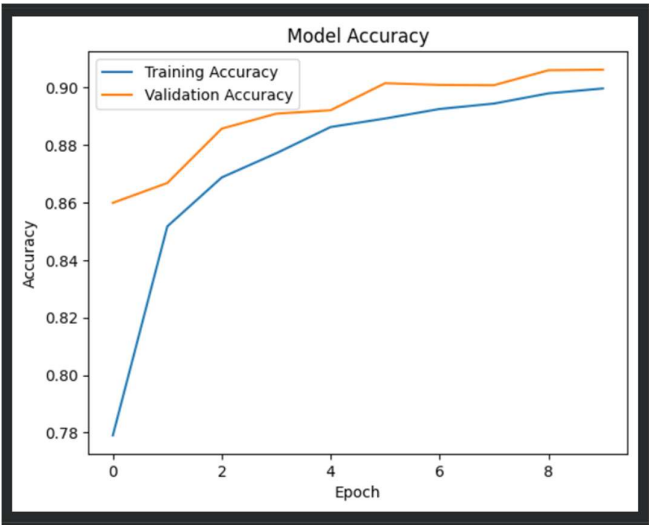
- **Support Vector Machine (SVM):** Linear kernel was used for classification.
- **K-Nearest Neighbors (KNN):** Number of neighbors $k = 5$.
- **Random Forest:** Ensemble model consisting of **100 decision trees**.
- **Principal Component Analysis (PCA):** Number of principal components = **340**.

These configurations were selected based on the methodology described in the research article and standard practices in machine learning model implementation.

6. Experimental Results and Comparative Analysis

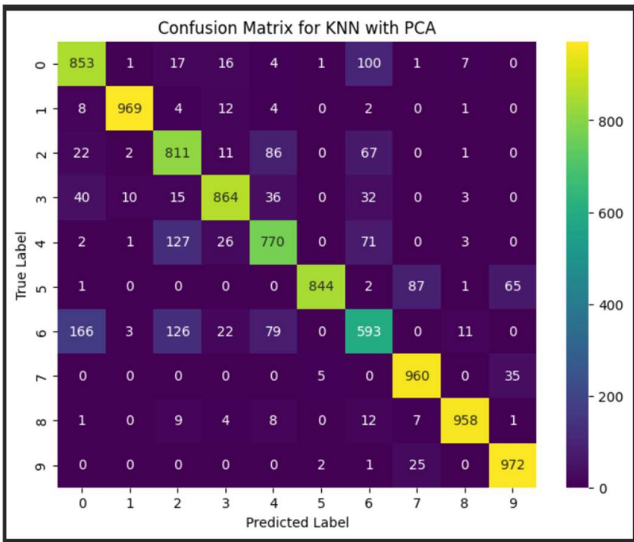
6.1 CNN Model Performance

The Convolutional Neural Network (CNN) model was trained using the Fashion-MNIST training dataset for **10 epochs**. The model achieved a **test accuracy of 90.62%**, demonstrating strong performance for image classification. The training and validation accuracy curves indicate that the model gradually improved during training while maintaining stable validation performance. The CNN architecture was able to automatically learn spatial features from the images, which contributed to its higher classification accuracy compared to traditional machine learning models.



6.2 Confusion Matrix Analysis

To further analyze the classification performance, a confusion matrix was generated for the **K-Nearest Neighbors (KNN) model with PCA**.

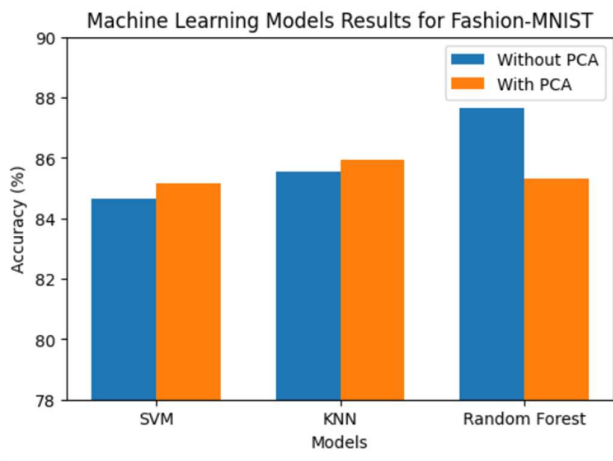


The confusion matrix illustrates how well the classifier predicts each clothing category. The majority of predictions appear along the diagonal of the matrix, indicating correct classifications. Some misclassifications occur between visually similar clothing categories such as **shirts, t-shirts, and pullovers**, which share similar visual patterns. This is a common challenge in fashion image classification tasks.

6.3 Machine Learning Model Comparison

To evaluate the impact of dimensionality reduction, three machine learning models were implemented and tested both **with and without PCA**. The obtained accuracy values are shown in below

Model	Without PCA	With PCA
SVM	84.64%	85.15%
KNN	85.54%	85.94%
Random Forest	87.64%	85.31%



The results show that PCA slightly improved the performance of **SVM and KNN models**, while the Random Forest model performed better without PCA. This behavior occurs because Random Forest algorithms can effectively handle high-dimensional data and may lose some useful information when dimensionality reduction is applied. Overall, the Random Forest model achieved the highest accuracy among the traditional machine learning models, while the CNN model achieved the best overall performance (90.62%) due to its ability to automatically extract meaningful image features.

7. Comparison with Research Paper and Discussion

In the research paper, the machine learning models such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest were evaluated on the Fashion-MNIST

dataset both with and without the application of Principal Component Analysis (PCA). The paper reported that dimensionality reduction using PCA slightly improved the performance of some models while reducing computational complexity. The results obtained in this implementation show a similar trend. The SVM classifier achieved an accuracy of **84.64% without PCA** and **85.15% with PCA**, indicating a small improvement after dimensionality reduction. Additionally, the execution time of the SVM model decreased significantly when PCA was applied, which demonstrates the efficiency gained by reducing the number of features.

The K-Nearest Neighbors model achieved an accuracy of **85.54% without PCA** and **85.94% with PCA**, which also indicates a slight improvement after applying PCA. This behavior is consistent with the observations reported in the research article. The Random Forest model achieved **87.64% accuracy without PCA**, which was higher than the performance obtained after applying PCA (**85.31%**). This result suggests that Random Forest algorithms are capable of handling high-dimensional data effectively and may lose useful information when dimensionality reduction is applied.

The deep learning model implemented in this study achieved a **CNN test accuracy of 90.62%**, which is very close to the results reported in the research article. This confirms that deep learning models, particularly convolutional neural networks, are highly effective for image classification tasks because they automatically learn spatial features from image data. Overall, the obtained experimental results are consistent with the trends reported in the selected research paper. Minor differences in accuracy may occur due to variations in implementation details, parameter settings, and computational environments.

8. Conclusion

This implementation used Fashion-MNIST dataset to train and evaluate several models, including Support Vector Machine, K-Nearest Neighbors, Random Forest, and Convolutional Neural Networks.

The results demonstrated that traditional machine learning models can achieve good classification performance when applied to flattened image features. Among the machine learning models, the Random Forest classifier achieved the highest accuracy. The application of Principal Component Analysis helped reduce the dimensionality of the data and improved computational efficiency for some models. The convolutional neural network achieved the highest overall accuracy of **90.62%**, highlighting the advantage of deep learning models in extracting meaningful spatial features from images. These results confirm the effectiveness of CNN architectures for image classification tasks.

The implementation results closely align with those reported in the selected research article, indicating that the methodology was successfully reproduced. Future work may explore more advanced deep learning architectures and transfer learning models to further improve classification performance.